Sneha Mondal

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Summary

Data engineer with 3+ years building and running production data platforms end to end: API ingestion, orchestration, warehouse modeling, transformation as code and data quality, as lead data engineer on a six-tenant platform on Google Cloud. Owned it in production, including monitoring, root-cause debugging, reliability work and a **96%** cut in monthly cloud spend.

Experience

Lead Analytics Engineer (Data Engineering) Jan 2025 – Sep 2026
YNK Incubator (3PL and umbrella company for Veiled Collection + DTC brands) *Elmwood Park, NJ*

- Built and owned the multi-tenant data platform for **six** DTC brands: event-driven ingestion on Cloud Run, Pub/Sub, Eventarc and Cloud Tasks into a BigQuery warehouse per client, with tenant separation enforced at the IAM layer (per-tenant service accounts, least-privilege roles, scoped delegation) on a shared base with brand-specific logic.
- Orchestrated scheduled and event-driven pipelines with dependency ordering, backoff retries and checkpointed backfills that resumed after a failure. Modeled transformations as version-controlled SQL in Dataform (dependency graph, assertions, incremental models, CI/CD), so every change was tested before production.
- Grew Sheets and Apps Script tooling into a client portal (Vercel, Neon Postgres) with contract-first, fail-loud validation so bad data failed at the boundary; trained the new team member on how the platform and the APIs worked.

Lead Data & Analytics Engineer Feb 2023 – Sep 2026
Veiled Collection (same employer group, scope expanded to the YNK platform above in 2025) *Clifton, NJ*

- Built the original single-brand platform (Shopify API into append-only tables, MERGE into current-state models) and cut operational data errors **95%** across **44K SKUs** with validation gates, daily snapshots, snapshot-diff reconciliation and alerting, so buying decisions ran on counts that tied back to source. Reconciled fulfillment status across Shopify, ShipStation and the warehouse, flagging stalled orders.
- Built the Buy Plan forecasting service: sell-out dates, reorder points and **6-month** order quantities by size and color, which the buying team planned purchasing against, validated against actual sell-through and corrected where projections drifted.
- Diagnosed a runaway Eventarc to Cloud Run feedback loop and cut monthly GCP spend **96%**, roughly **$8,000 to $300**: event-payload gating, deterministic task names against duplicate work, minimum instances to zero, partitioning and clustering.

Technical Skills

Languages	Python (Pandas, NumPy, pytest), SQL (CTEs, window functions), JavaScript/TypeScript, R, Java
Cloud & warehouse	GCP: BigQuery (3 yrs production, partitioning, clustering, cost tuning), Cloud Run, Cloud Functions, Pub/Sub, Eventarc, Cloud Tasks, Scheduler. Snowflake (project scale: loading, MERGE, least-privilege roles). PostgreSQL (Neon), MySQL, SQL Server, MongoDB
Transformation	Dataform in production (version-controlled SQL, dependency graph, assertions, incremental materializations, CI/CD); Kafka, dbt, DuckDB and Airflow at project scale; dependency ordering, backoff retries, checkpointed backfills, idempotent replay; Docker, Git, CI/CD
Quality & APIs	Data contracts, fail-loud validation, snapshot-diff reconciliation, freshness and row-count checks, monitoring and alerting, lineage; REST, GraphQL, Shopify Admin API, webhooks, ShipStation, pagination, rate limits
Analytics	Looker Studio, Tableau; dimensional modeling, SCD Type 2; scikit-learn, NLP; Claude Code and Cursor

Projects

- **Payments Reconciliation Copilot** github.com/snehamondal-data/payments-recon-copilot
  Kafka, Snowflake, dbt + MetricFlow, Airflow, Anthropic API (**328** tests). Streaming reconciliation of synthetic payments data with idempotent replay (**0** duplicate rows), dbt matching that found all **30** planted exceptions, identical on Snowflake and DuckDB, and a daily Airflow DAG; an AI copilot (read-only Snowflake role) proposes fixes for human approval, with **0** unsafe proposals on 35 planted cases under code-enforced escalation rules, plus a semantic layer for plain-English questions.
- **Data Reliability Pipeline & Analytics Agent** github.com/snehamondal-data/marketing-data-reliability-pipeline
  Python, dbt, DuckDB, Anthropic API. Idempotent incremental ingestion into multi-tenant marketing and order marts in dbt: **84** dbt tests per tenant, **83** pytest. An LLM agent over a semantic layer adds parsed SELECT-only guardrails (tenant-scoped, row and turn caps, logged), deterministic fallbacks and a 14-question eval harness, **10/10** live on the original 10.
- **Trickle-Down Trend Tracker** github.com/snehamondal-data/fashion-trend-engine
  Python, pandas, DuckDB, pytest. Tested whether FW26 runway trends trickle down to mid-price and mass brands using news coverage for **14** trends; the data contradicted the hypothesis and the write-up says so. Built a buy-planning tool that turns press momentum into order adjustments, validated on a simulated store with planted delays and a null test; **117** tests.

Education

Rochester Institute of Technology, M.S. Data Science, 2025 **University of Calcutta**, B.S. (Honors) Mathematics, 2021